ONI

Nanoimager

A benchtop solution for single-molecule imaging

ONI

TECHNICAL SPECIFICATIONS

Nanoimager

Resolution	**Single-molecule localization microscopy (SMLM)** 20 nm in XY (incl. STORM, DNA-PAINT, PALM)* **Widefield/TIRF** Measured as FWHM of fluorescent signal (488 nm): 236 +/- 16 nm
Illumination Control	Option to select illumination angle: 0°-65° Epifluorescence/widefield | HILO | TIRF | LED array for bright-field imaging Flat-field homogeneous laser illumination: coefficient of variation <22%
Lasers	SMLM in multiple colors (4 lasers) and 2 simultaneous channels (dichroic mirror split at 640 nm) **Power** At sample: 405 nm (15 mW), 488 nm (250 mW), 561 nm (250 mW), 640 nm (250 mW) At source: 405 nm (1 W), 488 nm (1 W), 561 nm (750 mW), 640 nm (1 W) **Switch on time** 405 nm (1 ms), 488 nm (1 ms), 561 nm (3 ms), 640 nm (1 ms)
Power Density	4 kW/cm^2 (0.24 kW/cm^2 for UV) Continuous laser
Temperature Control	Running temp 30-32°C Heating up to 42°C 1.5 - 2 hours warm up Sample temperature stable to +/- 0.1°C at steady state +/- 0.2°C with typical light program engaged** +/- 0.3°C with 100% laser power Precise temperature control, heating elements keep the entire unit at a uniform temperature. Ideal for live cell imaging and single-particle tracking
Focus Z-lock	Controllable Z offset +/- 10 µm from coverslip interface Rapid and precise stabilization of microscope focus, with locking technology to minimize Z-drift

Channel Mapping	Sub-pixel alignment between optical channels
Mechanical Stability	<1 µm/K drift
Product Dimensions	**Nanoimager** 21.5 cm (w) x 21.5 cm (d) x 15.5 cm (h) **Light Engine** 21.5 cm (w) x 42.0 cm (d) x 45.0 cm (h)
Imaging Technologies	**2D and 3D single-molecule localization microscopy (SMLM)** dSTORM, PALM, DNA-PAINT **Single-particle tracking (SPT)** **Total internal reflection fluorescence (TIRF)** TIRF achieved by objective-based TIRF Microscopy TIRF illumination depth: 488 nm (216 nm), 561 nm (248 nm), 640 nm (283 nm) TIRF depth is dependent on the incident wavelength, incident angle, refractive index of coverslip and refractive index of sample*** **Förster resonance energy transfer (FRET)**
Camera and Temporal Resolution	**Camera** Hamamatsu Orca Flash4.0 v3 **Frame rate** Full FOV: up to 10 ms / 100 Hz Cropped FOV: up to 1 ms / 1000 Hz **RMS read noise** 1.6 e-
Objectives & Field of View	**Objective compatibility** Standard: 100X NA 1.45**** **FOV with 100X objective** 50 µm x 80 µm **Sample stage movement** 17.5 x 17.5 x 5 mm XYZ travel range

*Better resolution may be possible

**Typical light program uses lasers at 100% for 40 min, followed by 20 min with lasers off, repeated over several hours with system heating set to 32°C

***TIRF: The incident angle is taken above the critical angle, 62 degrees, which is dependent on the refractive index of coverslip and sample. Refractive index of the coverslip is 1.52 and sample is 1.33, as for many of our use cases

****Please consult with ONI for different objective compatibility options

Interested in learning more?

@ONI @oniHQ @oni_bio @oni-bio

ONI, Linacre House, Jordan Hill Road, Oxford, OX2 8TA, UK
ONI Inc. 11045 Roselle St. Suite 120, San Diego, CA 92121, USA